

Receiver-Guided Reinforcement Learning-based Congestion Control for IoT

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Abstract—The abstract goes here.

Index Terms—IoT, TCP, congestion control, reinforcement learning.

I. INTRODUCTION

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II. RELATED WORKS

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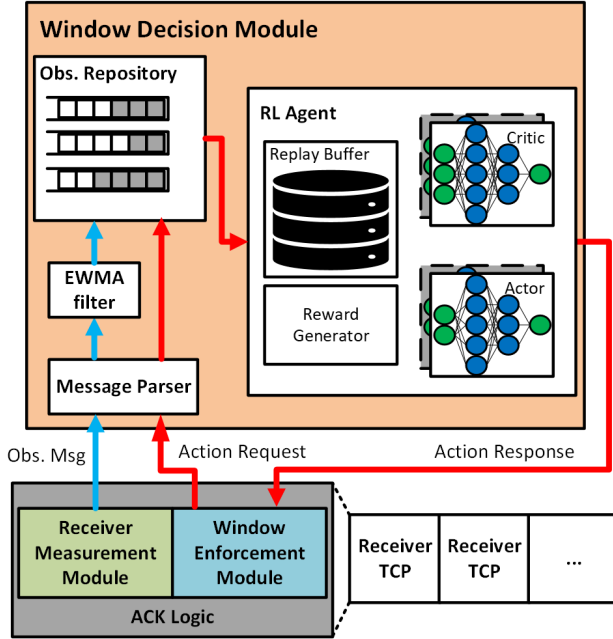


Fig. 1. System architecture of our proposal.

III. SYSTEM DESIGN

Our proposal contains three modules in the receiver side for adjusting sender $cwnd$, Receiver Measurement Module, Window Decision Module, and Window Enforcement Module. For better compatibility of our design, TCP sockets in both sender and receiver are extended or modified for adding interfaces and able to send three control messages to achieved our features.

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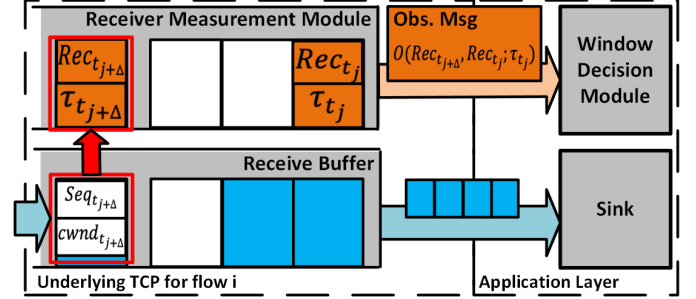


Fig. 2. An example of estimation task completion of flow i .

A. Control Message types definition

The receiver TCP socket is extended to support communication with the RL agent. Three types of messages are defined:

- **Observation Message:** carries the observed network status and is sent to the agent for storage in the replay buffer.
- **Action Request:** triggers the agent to infer and return a control action based on the current state.
- **Connection Close:** notifies the agent that a flow has terminated, allowing it to clear the corresponding internal states.

B. Receiver Measurement Module

Our scheme relies on reliable state observations to ensure stable training. Therefore, the states in our design are required to reflect the congestion extent while remaining observable at the receiver. Despite RTT is a fine-grained and valuable indicator of congestion, it is difficult to directly observe in the receiver side.

To address the limitation above, we derive a receiver-side congestion extent metric based on the $cwnd$ elapsed time ($t_{elapsed}$). We extend the sender TCP feature to include its $cwnd$ in TCP header. In standard TCP congestion avoidance, the congestion window $cwnd$ is increased by 1 MSS after an entire $cwnd$ bytes has been acknowledged. This nature results in an approximate growth rate of 1 MSS per RTT, in which the method is named Additive Increment.

We utilise Additive Increment growing feature to enable the receiver to estimate network congestion extent by calculating the elapsed time for a flow to complete its $cwnd$. In particular, upon every packet receive event that includes a $cwnd$ value different from the last value received, the Receiver Measurement Module establishes a new estimation task.

Algorithm 1 Receiver Measurement and Observation Generation

Input: Flow Id i , Time t_j , Arrived Packet p_{i,t_j} , Flow Observation Queue q_i

Output: Observation o_{i,t_j} , Control Message M

- 1: Extract Seq_{i,t_j} , $cwnd_{i,t_j}$, and $RxBytes_{i,t_j}$ from p_{i,t_j}
 - 2: $\tau_{i,t_j} = Seq_{i,t_j} + cwnd_{i,t_j}$
 - 3: Obtain $Losses_{i,t_j}$ from $TcpSocket_i$
 - 4: $B_i(t_j) = B_i(t_{j-1}) + RxBytes_{i,t_j}$
 - 5: $L_i(t_j) = L_i(t_{j-1}) + Losses_{i,t_j}$
 - 6: $Rec_{i,t_j} = [B_i(t_j), L_i(t_j), t_j]$
 - 7: Obtain Rec_i^{head} and τ_i^{head} from the head of q_i
 - 8: **if** $Seq_{i,t_j} \geq \tau_i^{head}$ **then**
 - 9: Dequeue Rec_i^{head} and τ_i^{head} from q_i
 - 10: $o_{i,t_j} = O(Rec_{i,t_j}, Rec_i^{head})$
 - 11: Include o_{i,t_j} into M
 - 12: Send M to agent
 - 13: **else if** $cwnd_{i,t_j} \neq cwnd_{i,t_{j-1}}$ **then**
 - 14: Enqueue Rec_{i,t_j} and τ_{i,t_j} into q_i
 - 15: **end if**
-

A new task is initialised with a record (Rec) for the current flow state and a threshold (τ), e.g., for flow i started at time t_0 and estimated at time t_j , defined as below:

$$Rec_{i,t_j} \triangleq [B_i(t_j), L_i(t_j), t_j] \quad (1)$$

where we set τ to be the sum of sequence number (Seq_{i,t_j}) and the $cwnd$ value from its header. We also maintain a cumulative received bytes ($RxBytes_{i,t}$) and cumulative packet losses ($Losses_{i,t}$) counter in the receiver TCP, defined as follows:

$$\tau_{i,t_j} = Seq_{i,t_j} + cwnd_{i,t_j} \quad (2)$$

$$B_i(t_j) = \sum_{n=t_0}^{t_j} RxBytes_{i,n} \quad (3)$$

$$L_i(t_j) = \sum_{n=t_0}^{t_j} Losses_{i,n} \quad (4)$$

An estimation task is completed at $t_{j+\Delta}$ when a coming packet is seen with a sequence number satisfying $Seq_{i,t_{j+\Delta}} > \tau_{i,t_j}$. We calculate this task duration as the $cwnd$ elapsed time i.e., $t_{elapsed} = t_{j+\Delta} - t_j$. We then calculate the goodput (g_{i,t_j}) and packet losses during the estimation task, where $g_{i,t_j} = \frac{B_i(t_{j+\Delta}) - B_i(t_j)}{t_{elapsed}}$, and $\ell_{i,t_j} = L_i(t_{j+\Delta}) - L_i(t_j)$. Therefore, an observation is formed denoted as $o_{i,t_j} = O(Rec_{i,t_{j+\Delta}}, Rec_{i,t_j}; \tau_{i,t_j})$, with the observation as the result of the generation process defined as follows:

$$O(Rec_{i,t_{j+\Delta}}, Rec_{i,t_j}; \tau_{i,t_j}) = [g_{i,t_j}, \ell_{i,t_j}, t_{elapsed}]$$

The eventual observation is encapsulated into a Observation Message and sent to the Window Decision Module for storage. The consecutive process triggered by packet received event can be seen in Algorithm 1.

Algorithm 2 Window Decision Module

Input: Control Message M , Replay Buffer B , Flow State Buffer B_i for flow i , Batch Size N

Output:

- 1: **if** $M = \text{"Connection Close"}$ **then**
 - 2: Clear B_i
 - 3: **else if** $M = \text{"State"}$ **then**
 - 4: Extract s_i from M
 - 5: Store s_i into B_i
 - 6: **else if** $M = \text{"Action Request"}$ **then**
 - 7: Compute the average state $\bar{s}_i$ over B_i
 - 8: $a = \mu(\bar{s}_i; \theta^\mu)$
 - 9: Store transition $(\bar{s}_i, a_i, r_i, \bar{s}_i')$ into B
 - 10: Randomly sample N transition from B
 - 11: Compute $y_t = r(s_t, a_t) + \gamma Q^-(s_{t+1}, \mu^-(s_{t+1}))$
 - 12: Compute $L(\theta^Q) = \mathbb{E}[(Q(s_t, a_t; \theta^Q) - y_t)^2]$
 - 13: Update Critic minimising $L(\theta^Q)$
 - 14: Compute $\nabla_{\theta^\mu} J(\theta^\mu) = \mathbb{E}[\nabla_{a_t} Q(s_t, \mu(s_t)) \nabla_{\theta^\mu} \mu(s_t)]$
 - 15: Update Actor via gradient ascent $\nabla_{\theta^\mu} J(\theta^\mu)$
 - 16: $\theta^{\mu^-} \leftarrow \tau \times \theta + (1 - \tau) \times \theta^{\mu^-}$
 - 17: $\theta^{Q^-} \leftarrow \tau \times \theta + (1 - \tau) \times \theta^{Q^-}$
 - 18: **end if**
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C. Window Decision Module

Window Decision Module consists of a Reinforcement Learning model to output a scaling factor to underlying TCP socket periodically triggered by Action Request from Window Enforcement Module. An enforced congestion window is then calculated with the factor by the underlying TCP socket and included in ACK to the sender. The sender will overwrite its $cwnd$ by the enforced congestion window. Therefore, to implement a continuous control to modified TCP, we choose Deep Deterministic Policy Gradient (DDPG) as agent in our design.

1) *State Generation*: An observation is generated on the packet arrival event while the action is requested periodically. In our design, the action period is set to long enough to ensure we have multiple observations stored by an Observation Repository. Therefore, we first use an exponential weighted moving average (EWMA) filter to smooth the observations before storing them in the repository. This is done to mitigate the noise when the observation is generated in a varying network environment e.g., Wi-Fi. After storing, when an action request is received, the average of the stored observations during this action period is calculated as input to our agent, seen as follows when n observations are stored:

$$s_t = \left[\frac{1}{n} \sum_{i=1}^n g_i, \frac{1}{n} \sum_{i=1}^n t_{elapsed,i} \right] \quad (5)$$

2) *Reward Design*: For improving network performance for each flow, the first component of our reward is based on network power metric to reflect individual flow performance, defined as follows:

$$r_{1,t} = \frac{\frac{1}{n} \sum_{i=1}^n g_i}{\frac{1}{n} \sum_{i=1}^n t_{elapsed,i}}$$

We also take the advantage of the global performance store in repository to design the second component for improving fairness among flows. Jain's fairness index is introduced considering N flows defined as follows:

$$r_{2,t} = \frac{\left(\sum_{i=1}^N \bar{g}_i\right)^2}{N \sum_{i=1}^N \bar{g}_i^2}$$

Eventually, we design our reward as:

$$r_t = \begin{cases} -2, & \text{if } \ell_t > 0 \\ r_{1,t} - (1 - r_{2,t})^2, & \text{otherwise} \end{cases} \quad (6)$$

3) *Training*: Our agent consists of an actor network determining a deterministic action $a = \mu(s; \theta^\mu)$ parameterised by θ^μ and a critic network to approximate the expected future return, defined as $Q(s, a; \theta^Q) = \mathbb{E}[R_t | s_t, a_t]$ with parameters θ^Q . The return R_t is the sum of discounted future return, i.e., $R_t = \sum_{i=t}^T \gamma^{i-t} r(s_i, a_i)$. Approaching to the approximation is done by updating the critic network through minimising the mean square error between its output and a target function (y_t), shown as follows:

$$L(\theta^Q) = \mathbb{E}[(Q(s_t, a_t; \theta^Q) - y_t)^2]$$

$$y_t = r(s_t, a_t) + \gamma Q^-(s_{t+1}, \mu^-(s_{t+1}; \theta^{\mu^-}); \theta^{Q^-})$$

where $Q^-(s, a; \theta^-)$ and $\mu^-(s; \theta^{\mu^-})$ are target critic network and target policy network. Their parameters are updated by calculating the exponentially weighted moving average (EWMA) of critic network and policy network, i.e., $\theta^- \leftarrow \tau \times \theta + (1 - \tau) \times \theta^-$ where the τ is the weight parameter. Subsequently, with the updated critic network, policy performance of current actor network is predicted, expressed as $J(\theta^\mu) = \mathbb{E}[Q(s_t, \mu(s_t; \theta^\mu))]$. Training an actor network is done by updating its parameters through policy performance gradient ascent. Specifically, by applying the chain rule, the gradient of the policy performance, is computed as:

$$\nabla_{\theta^\mu} J(\theta^\mu) = \mathbb{E}[\nabla_{a_t} Q(s_t, \mu(s_t; \theta^\mu)) \nabla_{\theta^\mu} \mu(s_t; \theta^\mu)]$$

D. Window Enforcement Module

For an RL-controlled CC, it is important that waiting agent for action generation should not block the TCP normal behaviour. Therefore, we decouple the Window Decision Module, Receiver Measurement Module and standard feature to ensure none of them will block any other. Therefore, an ACK response will not be delayed in our scheme. In our design, an action request message will be sent to agent periodically by Window Decision Module. The resulting action is a factor within $[0, 1]$. Once the Window Decision Module receive the action, it will extract the latest $cwnd$ in Receiver Measurement Module and compute the enforced congestion window ($cwnd_\mu$) using the following:

$$cwnd_\mu = cwnd \times scaling_factor^a, \quad n \in [0, 1]$$

Then $cwnd_\mu$ will be included into the ACK header and take effect at the sender side.

IV. SIMULATION SETUP AND RESULTS

V. CONCLUSION

The conclusion goes here.

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